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Executive Summary

This document describes the GDAM Green Hydrogen LCOH Explorer, an open-source interactive dashboard for green hydrogen (GH2) project developers in India. Its purpose is to help a developer determine the levelised cost of hydrogen (LCOH) from an alkaline electrolyser procuring power exclusively from the IEX Green Day-Ahead Market (GDAM), with no captive renewable plant and no battery storage.

The tool ingests four years of real 15-minute GDAM block prices (Aug 2022 – Aug 2026) and computes LCOH as a function of a single operational choice: the maximum landed power price the developer is willing to pay. Everything else follows from the price distribution and the cost stack.

Key finding	Value
Best available LCOH (Gujarat, AY2025-26, 20-yr)	Rs 268/kg
Best available LCOH (Gujarat, AY2025-26, 7-yr)	Rs 309/kg
Best available LCOH (Rajasthan, AY2025-26, 20-yr)	Rs 278/kg
Grey hydrogen benchmark	Rs 250/kg
Optimal capacity factor (all years)	64 – 69%
Optimal bid ceiling (Gujarat, AY2025-26)	Rs 4.98/unit landed (Rs 4.43 MCP)
Required plant size for 500 TPA at optimum	4.60 MW
Stack replacement year (at optimum)	Year 10.6
Rajasthan cost penalty vs Gujarat	Rs 10–12/kg (charge structure only)

■ **IMPORTANT** — This tool is intended as a first-pass scoping instrument for the GH2 producer. It uses a single representative year of exchange prices as a proxy for long-run power cost. The low-price regime observed from 2024 onward is approximately 2.5 years old. A 20-year investment must not be underwritten solely on 2.5 years of history. Excluded costs (electricity duty, STU demand charges, GST, land, insurance, working capital) are material and listed in Section 12.

1. What This Tool Does

An alkaline electrolyser is connected to the GDAM. No captive solar or wind plant, no battery. The developer bids into the exchange every day and buys power only when the 15-minute block price is at or below a chosen ceiling. The electrolyser runs at full load whenever a qualifying block is bought, subject to a minimum continuous run to protect equipment (typically 2 hours for alkaline).

For any given ceiling, the historical price data yields an annual operating-hour count and a volume-weighted average power cost. These, combined with the developer's cost stack, produce the LCOH. The dashboard sweeps the ceiling from the year's minimum price to its maximum and plots the resulting LCOH curve. The curve has a characteristic U-shape: too low a ceiling means too few hours, driving up unit fixed costs; too high a ceiling means buying expensive evening peak power, driving up the power component.

The developer's decision variable is the ceiling. Everything else is an output.

2. Data

2.1 Source

IEX Green Day-Ahead Market (GDAM), 15-minute block prices. The raw file covers 1 April 2022 to 21 August 2026, 153,984 blocks. Two fields are spliced: *MCP (Rs/MWh)* is used to 20 February 2024, and *unconstrained_m_c_p* from 21 February 2024 onward. Both are converted to Rs/kWh by dividing by 1,000. Unconstrained MCP is used throughout; 679 of 87,648 flagged blocks (0.77%) were congested — too small a share to move a volume-weighted average and excluded consistently.

2.2 Price years

Four complete August-to-August windows are used. Calendar years are deliberately avoided because they truncate the seasonal cycle: the Jan–Mar dead zone (zero hours below Rs 1.5 MCP) and the Sep–Oct cheap window are the two most important seasonal features, and a calendar year boundary splits at least one of them.

Year	Window	Days	Mean	p10	p25	Median
AY2022-23	22 Aug 22 – 21 Aug 23	364	5.80	3.40	3.93	4.89
AY2023-24	22 Aug 23 – 21 Aug 24	365	5.73	3.05	3.89	4.54
AY2024-25	22 Aug 24 – 21 Aug 25	364	4.61	2.10	3.00	4.00
AY2025-26	22 Aug 25 – 21 Aug 26	364	4.61	1.63	2.60	3.67

Year	h below Rs 1.5	h below Rs 2.5	h below Rs 3.5
AY2022-23	0	43	1,030
AY2023-24	23	359	1,366
AY2024-25	338	1,321	3,248
AY2025-26	676	2,059	3,761

Table 1: GDAM price statistics by Aug–Aug year. All prices in Rs/kWh; h values in hours per year.

AY2022-23 and AY2023-24 show near-zero hours below Rs 1.5 MCP. Any LCOH result from these years should be read as a historical benchmark, not a forward projection. The low-price regime — driven by growing solar surplus — is a post-2024 phenomenon.

3. Landed Cost Formula by State

Open-access GH2 producers qualify for concessions under the Ministry of Power's Green Hydrogen Open Access notification. The landed cost is exchange MCP plus residual transmission and wheeling charges, times a composite loss factor:

State	Formula	Adder (Rs/unit)	Loss factor	Basis
Gujarat	$\text{landed} = (\text{MCP} + 0.250) \times 1.065$	0.250	1.065	50% wheeling waiver; ISTS charges waived for 25 yr
Rajasthan	$\text{landed} = (\text{MCP} + 0.365) \times 1.088$	0.365	1.088	50% wheeling waiver; ISTS charges waived for 25 yr

Table 2: State landed-cost formulas.

The Rs 10–12/kg Rajasthan cost penalty relative to Gujarat is entirely attributable to the higher residual charge stack. The exchange price is identical for both states — GDAM is a single national market clearing at one price per block.

■ The ISTS waiver applies only to projects commissioned before 31 December 2030. Electricity duty and STU demand charges are NOT included in these formulas and are excluded from all LCOH calculations in this tool. See Section 12.

Why the sweep is in MCP space. Both formulas are monotonic affine transforms of MCP ($\text{landed} = a \cdot \text{MCP} + b$, where a and b are positive constants). Affine monotonic maps preserve both ordering and arithmetic means. Consequently: (i) the set of blocks selected at a landed ceiling C equals the set selected at the MCP threshold $t = C/a - b/a$; (ii) the volume-weighted mean landed cost of selected blocks is $a \cdot w + b$, where w is the volume-weighted mean MCP. The browser therefore looks up the pre-computed MCP sweep and applies the state transform client-side. One sweep serves every state.

4. Dispatch Logic — the Min-Run Rule

GDAM is a day-ahead market. Block prices are published the previous evening, so the dispatch schedule is executable without forecast error. No DSM penalties or imbalance charges arise from GDAM trading.

Pure on/off dispatch at block level does not suit alkaline electrolyzers: cold-start requires a thermal ramp, and sub-15-minute shutdowns accumulate cell degradation. The tool imposes a **minimum continuous run** (default 2 hours = 8 blocks):

Algorithm

Step 1 — Qualify blocks: all blocks with landed price $\leq$ ceiling are candidates.

Step 2 — Identify contiguous runs of qualified blocks.

Step 3 — Runs of $\geq \text{min_run}$ blocks pass through unchanged.

Step 4 — Shorter runs are extended outward, one block at a time, always taking the cheaper of the two adjacent blocks (left or right), until the run reaches min_run blocks.

Step 5 — The extended window is kept ONLY if its mean landed price $\leq$ ceiling. Otherwise the entire window is discarded.

The penalty at the optimum is Rs 0.0–0.4/kg. At the optimal 64–69% capacity factor, the electrolyser runs two-thirds of all blocks. Cheap periods are multi-hour solar-surplus windows and overnight troughs, not scattered single blocks. The minimum-run filter therefore almost never binds at the LCOH optimum. It would bind hard on a very low ceiling strategy (< 5% CF), providing an additional argument against such designs.

5. LCOH Formula — Full Derivation

Step 1 — Lookup operating hours and weighted MCP

```
t = ceiling_landed / loss_factor - adder
hours, w = interpolate(sweep, t) ← linear interpolation, never extrapolated
landed = (w + adder) × loss_factor
```

Step 2 — Production

```
annual_kg = MW × 1000 × hours × load_factor ÷ SEC
CF = hours ÷ 8760
```

Step 3 — Capital cost annualisation

```
CRF(r, N) = r(1+r)^N / ((1+r)^N - 1)
annualised_capex = (capex - capex × residual / (1+r)^N) × CRF(r, N)
```

Step 4 — Stack replacement sinking fund

```
stack_cost = MW × 1000 × ELY_kw × stack_pct
interval = stack_life_hours ÷ hours
stack_sinking_fund = CRF(r,N) × Σ stack_cost / (1+r)^(k×interval)
for k = 1, 2, 3, ... while k×interval < N
```

Step 5 — Total LCOH

```
LCOH = (annualised_capex + annual_O&M; + stack_sinking_fund) ÷ annual_kg
+ SEC × landed
+ water_cost_per_kg
```

Reference case — Gujarat, AY2025-26, 20-year recovery:

Component	Calculation	Rs/kg	% of LCOH
Capital recovery	$285183990 \times 0.117460 \div 500,000$	67.00	25.0%
Fixed O&M;	$285183990 \times 0.04 \div 500,000$	22.81	8.5%
Stack replacement	$PV = 26093399;$ $\times 0.117460 \div 500,000$	6.13	2.3%
Power (landed)	$52 \times (2.843 + 0.250) \times 1.065 =$ 52×3.2940	171.29	63.9%
Water	$15 \text{ L/kg} \times \text{Rs } 60/\text{kL} \div 1,000$	0.90	0.3%
LCOH Total		268.13	100.0%

Table 3: LCOH breakdown. Design: 4.599 MW, 5,654 h/yr, CF 64.5%, capex Rs 28.52 cr. $CRF(10\%, 20) = 0.117460$.

6. Capex Build-Up

All uplifts are applied to the base electrolyser installed cost. The same Rs 62,010/kW total is the default across both states; the developer can override any line item in the Advanced panel.

Item	Uplift (%)	Rs/kW	Running total
Electrolyser installed (equipment + BOP + installation)	—	39,000	39,000
Civil works / site prep	12%	4,680	43,680
Water treatment (DM/RO)	6%	2,340	46,020
Power evacuation, HT yard, transformer	10%	3,900	49,920
H2 purification, drying, compression	10%	3,900	53,820
Instrumentation, safety, fire, controls	6%	2,340	56,160
Owner cost, EPC margin, contingency	15%	5,850	62,010
All-in plant capex	+59%	62,010	62,010

Table 4: Capex build-up. All uplifts are on the base electrolyser installed cost of Rs 39,000/kW.

■ These are indicative uplifts based on Indian EPC norms for MW-scale projects. They have NOT been validated against a project-specific bill of quantities. A real BOQ could move the all-in figure by $\pm 20\%$. Since capex recovery is 25–40% of total LCOH at the optimum, a 20% capex reduction saves roughly Rs 15–20/kg — enough to reach grey parity from the AY2025-26 base without any other intervention.

7. Reference Results by Price Year

Gujarat GH2 open access. 20-year capital recovery at 10% discount rate. Minimum run 2 hours. Default cost stack. AY2025-26 is the recommended base case — the only complete Aug-to-Aug year ending on the most recent data.

Year	Opt.ceil (landed)	Opt.ceil (MCP)	Wtd landed	Hours	CF	MW(500TP A)	Capex(cr)	LCOH
AY2022-23	6.39	5.95	4.76	5,756	65.7 %	4.52	28.05	342.8
AY2023-24	6.18	5.75	4.50	5,660	64.6 %	4.59	28.49	330.7
AY2024-25	5.32	4.75	3.71	6,000	68.5 %	4.33	26.89	284.8
AY2025-26	4.98	4.43	3.29	5,654	64.5 %	4.60	28.54	268.1

Table 5: Optima by year — Gujarat. MCP equiv. = ceiling_landed / 1.065 – 0.250.

The key structural finding: the optimal ceiling falls from Rs 6.39 to Rs 4.98 across four years — a 22% drop — while the optimal **capacity factor holds at 64–69%** throughout, and the plant size stays in a 4.33–4.60 MW band. The correct design rule is to size for approximately 65% CF and set the ceiling at whatever the 65th-percentile landed price of the representative year turns out to be. A fixed numerical ceiling is not portable across years.

For **Rajasthan**, add Rs 10–12/kg to each LCOH above. The Rs 10.2/kg differential (at AY2025-26) decomposes as: $(3.490 - 3.294) \times 52 = \text{Rs } 10.2/\text{kg}$ in power cost, with all other components unchanged.

8. Interpreting the U-Curve

The LCOH vs landed ceiling chart is the centrepiece of the dashboard. Understanding its shape prevents two common misinterpretations.

8.1 Why the curve rises at low ceilings

A lower ceiling means fewer qualifying hours. Fixed costs (capital recovery, O&M;) are spread over less production, so unit fixed cost rises steeply. At a Rs 1.5/unit ceiling in AY2025-26 the plant runs only 791 hours (9% CF) and capital recovery alone exceeds Rs 640/kg. The power component is very cheap (Rs 78/kg) but irrelevant — it is dominated by fixed costs.

8.2 Why the curve rises at high ceilings

A higher ceiling includes increasingly expensive evening and night blocks. The volume-weighted average landed power cost rises faster than fixed unit cost falls, because once the plant is running 70–80% of hours, most remaining blocks are expensive (Rs 6–10/unit). Buying these adds cost without proportionally reducing fixed unit cost.

8.3 The flat bottom

The optimum is shallow. In AY2025-26 Gujarat, the LCOH is within Rs 5/kg of its minimum across a Rs 1.5/unit ceiling band (roughly Rs 4.0–5.5). **Do not read three significant figures of precision into the optimal ceiling number.** The capacity factor at the optimum is the stable quantity; the ceiling is just the dial that produces it.

8.4 The two ceiling numbers

Every design has two distinct ceiling values that must not be confused:

- **Bid ceiling** — the number entered into the GDAM order: the landed price above which you stop buying. AY2025-26 Gujarat optimum: Rs 4.98/unit landed = Rs 4.43/unit MCP.
- **Weighted average landed cost** — the outcome that enters the power line of the LCOH. AY2025-26 Gujarat optimum: Rs 3.29/unit. The Rs 1.69/unit gap is large because most selected blocks cluster far below the marginal one (the skewed distribution has a long cheap tail).

8.5 Seasonality

At the AY2025-26 optimum (Rs 4.98 ceiling) the plant runs 48–81% utilisation by month. The weakest month is April (48%) — pre-monsoon, solar surplus not yet at maximum. The strongest months are September–November (75–81%) when solar surplus is large and peak demand has subsided. January–March run at 59–71%. **There is no dead season at the optimum** — the plant produces every month, ranging roughly 27–58 t/month against a 42 t/month average. Offtake agreements should be sized around the April minimum, not the January dead zone that applies only to ultra-low ceiling strategies.

9. The MW-Invariance Property

The LCOH-optimal ceiling, optimal operating hours, and capacity factor are mathematically independent of plant size (MW). This is not an approximation — it is an exact algebraic consequence of the cost structure.

Proof

```
annual_kg = MW × 1000 × hours / SEC ← proportional to MW
capex = MW × 1000 × capex_kW ← proportional to MW
annualised_capex / annual_kg = capex_kW × CRF(r,N) × SEC / (1000 × hours) ← MW
cancels
O&M; / annual_kg = capex_kW × om_pct × SEC / (1000 × hours) ← MW cancels
stack_sinking / annual_kg = ELY_kW × stack_pct × ... / hours ← MW cancels
power term = SEC × landed ← no MW
=> LCOH depends only on hours and landed_avg, not on MW.
```

Practical implication. The developer can independently choose plant size (to hit a target annual production) and operating strategy (ceiling, to hit a target LCOH). The dashboard shows a single U-curve that applies at any scale. MW scales nameplate capacity and actual output linearly, and scales capex linearly, but does not move the optimal point on the curve.

Verification. The self-test suite in the dashboard asserts that sweeping MW from 1 to 50 MW changes LCOH by less than Rs 0.01/kg and leaves the optimal ceiling and CF unchanged to four decimal places.

10. Three-Year Repricing

When the 3-year repricing checkbox is enabled, the dashboard computes a regulatory-style cost-of-service tariff with a declining rate base, allowing the GH2 price to be renegotiated every three years as capital is progressively recovered.

10.1 Mechanics

```
Year t:
if t = ceil(stack_life / hours):
rate_base += stack_cost
stack_dep = stack_cost / (plant_life - t + 1)
depreciation = capex / N (while t ≤ N, else 0)
return = r × opening_rate_base
capital_Rs_kg = (dep + stack_dep + return) / annual_kg

Block price (3-year blocks, levelised within block):
price_b = CRF(r, block_len) × Σ capital_Rs_kg_t / (1+r)^i
+ SEC × landed_avg + O&M;_per_kg + water
```

10.2 Block schedule (Gujarat, AY2025-26, 7-yr amortisation, 500 TPA)

Blo ck	Years	Opening rate base	Cap. charge	Contract price	Revenue/yr	vs grey
B1	1–3	Rs 28.52 cr	130.9	326	Rs 16.29 cr	1.30×
B2	4–6	Rs 16.30 cr	106.4	301	Rs 15.07 cr	1.21×

Blo ck	Years	Opening rate base	Cap. charge	Contract price	Revenue/yr	vs grey
B3	7–9	Rs 4.07 cr	32.8	228	Rs 11.39 cr	0.91×
B4	10–12	Rs 7.17 cr	17.8	213	Rs 10.64 cr	0.85×
B5	13–15	Rs 0.00 cr	24.5	219	Rs 10.97 cr	0.88×
B6	16–18	Rs 0.00 cr	20.2	215	Rs 10.76 cr	0.86×
B7	19–20	Rs 0.00 cr	16.5	212	Rs 10.58 cr	0.85×

Table 6: Repricing block schedule. Power component (Rs 171.3/kg) is the same in every block under TTM-constant assumption.

NPV-neutrality. Rolling 3-year repricing on a declining rate base is exactly NPV-neutral to both parties: the 20-year PV-levelised price to the offtaker equals the flat 20-year LCOH (Rs 268/kg). The repricing structure distributes the same total value across time without extracting rent from either side. The dashboard asserts this numerically in its self-test.

Commercial risk. Under 7-year amortisation the contract price falls from Rs 326 to Rs 228 at the second renewal — a 30% cut visible to the offtaker from day one. The developer should contract the *formula* (rate base + indexed power), not the schedule, and disclose the rate-base balance at each reset. The stack replacement causes a price step-up at block 5 (year 13–15), which should be pre-disclosed.

11. Independent Arithmetic Checks

The following checks were computed using a second, independent Python script (no imports from the main model) and verified against the dashboard's data.json. All results must match to within Rs 1.0/kg tolerance.

Check 1 — Capital Recovery Factor

$$\text{CRF}(10\%, 20) = 0.10 \times 1.10^{20} / (1.10^{20} - 1) = 0.117460 \checkmark$$

$$\text{CRF}(10\%, 7) = 0.10 \times 1.10^7 / (1.10^7 - 1) = 0.205405 \checkmark$$

$$\text{CRF}(10\%, 3) = 0.402115 \quad \text{CRF}(0\%, 10) = 1/10 = 0.100000 \checkmark$$

Check 2 — Capex Build-Up

$$39,000 \times (1 + 0.59) = 39,000 \times 1.59 = 62,010 \text{ Rs/kW} \checkmark$$

Check 3 — Plant Sizing for 500 TPA

$$\text{MW} = 500,000 \text{ kg} \times 52 \text{ kWh/kg} \div 5,654 \text{ h} \div 1,000 = 4.5985 \text{ MW} \checkmark$$

$$\text{capex} = 4598.51 \text{ kW} \times \text{Rs } 62,010 = \text{Rs } 28.5154 \text{ cr} \checkmark$$

Check 4 — LCOH Total (independent calculation)

$$\text{landed} = (2.843 + 0.250) \times 1.065 = 3.2940 \text{ Rs/unit}$$

$$\text{cap_rec} = 285183990 \times 0.117460 \div 500,000 = 66.995 \text{ Rs/kg}$$

$$\text{O\&M;} = 285183990 \times 0.04 \div 500,000 = 22.815 \text{ Rs/kg}$$

$$\text{stack} = 26093399 \times 0.117460 \div 500,000 = 6.130 \text{ Rs/kg}$$

$$\text{power} = 52 \times 3.2940 = 171.290 \text{ Rs/kg}$$

$$\text{water} = 0.900 \text{ Rs/kg}$$

$$\text{TOTAL} = 268.130 \text{ Rs/kg expected } 268.1 \text{ diff } +0.030 \checkmark$$

Check 5 — Stack Replacement Timing

$$60,000 \text{ h} \div 5,654 \text{ h/yr} = 10.6120 \text{ yr}$$

$$\text{First replacement at year } 10.61 - \text{inside } 20\text{-yr plant life, outside } 7\text{-yr recovery} \checkmark$$

Check 6 — Block 1 Repricing Consistency

$$\text{Block 1 capital charge} = 130.887 \text{ Rs/kg}$$

$$\text{Block 1 price} = 130.887 + 171.290 + 22.815 + 0.9 = 325.892 \text{ Rs/kg}$$

$$\text{Expected} \approx 325.9 \text{ diff } -0.008 \checkmark$$

Check 7 — MW Invariance (independent)

$$\text{MW} = 1: \text{LCOH} = 268.1200 \text{ diff from MW=5: } -0.000000 \checkmark$$

$$\text{MW} = 5: \text{LCOH} = 268.1200 \text{ diff from MW=5: } +0.000000 \checkmark$$

$$\text{MW} = 10: \text{LCOH} = 268.1200 \text{ diff from MW=5: } +0.000000 \checkmark$$

$$\text{MW} = 50: \text{LCOH} = 268.1200 \text{ diff from MW=5: } -0.000000 \checkmark$$

Check 8 — Gujarat vs Rajasthan Differential

$$\text{Raj landed} = (2.843 + 0.365) \times 1.088 = 3.4903 \text{ vs Guj } 3.2940$$

$$\text{Power diff} = (3.4903 - 3.2940) \times 52 = 10.21 \text{ Rs/kg expected } \approx 10 \checkmark$$

Case	Expected	Computed	Diff	Result
CRF(10%,20)	0.11746	0.12	-0.00	✓ PASS
CRF(10%,7)	0.205405	0.21	+0.00	✓ PASS

Case	Expected	Computed	Diff	Result
Capex/kW (Rs)	62010	62010.00	+0.00	✓ PASS
LCOH (Rs/kg)	268.1	268.13	+0.03	✓ PASS
Stack year	10.61	10.61	+0.00	✓ PASS
Block 1 price (Rs/kg)	325.9	325.89	-0.01	✓ PASS
Raj-Guj power diff (Rs/kg)	10.0	10.21	+0.21	✓ PASS
MW=1 LCOH = MW=5 LCOH	268.11997808 564126	268.12	-0.00	✓ PASS
MW=10 LCOH = MW=5 LCOH	268.11997808 564126	268.12	+0.00	✓ PASS
MW=50 LCOH = MW=5 LCOH	268.11997808 564126	268.12	-0.00	✓ PASS

Table 7: Independent arithmetic verification. All checks computed without importing the model module.

12. What This Tool Does NOT Model

The following cost items and structural features are excluded from the LCOH calculation. They are not minor. The developer must add them before using any output for investment decisions.

Excluded costs — material

Electricity duty: Varies by state; typically Rs 0.05–0.20/unit for industrial HT consumers. On 26,000 MWh/yr this is Rs 1.3–5.2 mn/yr directly to the power line.

STU demand charges: Applicable to EHV-connected buyers (~Rs 0.30–0.35/unit at high load factor, or a fixed Rs/kVA/month). At 64.5% CF this may add Rs 15–25/kg and is the largest single excluded item.

GST on power purchases: Input tax credit treatment depends on the GH2 end use; not universally available.

Working capital: Daily GDAM purchases require pre-deposited margin; roughly 5–10 days of power cost at peak purchase level.

Land and civil lease / acquisition: Excluded from the Rs 62,010/kW stack.

Insurance: Typically 0.3–0.5% of capex per year; Rs 0.85–1.4 cr/yr on the reference design.

Regulatory and statutory costs: Open access application fees, banking charges, green certificate compliance.

Excluded structural features

Storage / battery: The tool models exchange-only dispatch. Adding storage changes both the cost stack and the hour-selection logic fundamentally.

Captive RE plant or PPA: No wind, solar, or hybrid plant. Power is sourced entirely from GDAM.

Price escalation: All figures are real-terms, held at the chosen representative year. No inflation, no trending.

SIGHT subsidy (Component II): The electrolyser capex subsidy under SIGHT (if and when disbursed) would reduce the capital recovery line by Rs 10,000–15,000/kW if the scheme covers all-in costs.

States other than Gujarat and Rajasthan: The tool uses verified charge stacks for these two states only. Formulas for other states are not validated.

Plant degradation: Electrolyser output degradation over stack life (typically 0.5–1.0% per year of rated capacity) is not modelled.

Partial load operation: The model assumes full-load operation whenever the electrolyser runs. Alkaline units can turn down to ~20–40% of rated capacity; partial-load economics are excluded.

Carbon accounting: No carbon credit income or green premium pricing.

Data limitations

The low-price regime (cheap hours below Rs 1.5 MCP) is approximately 2.5 years old (visible from AY2024-25 onward). A 20-year investment should not be underwritten solely on 2.5 years of price history. In AY2022-23 and AY2023-24 there were essentially no hours below Rs 1.5, the LCOH minimum exceeded Rs 330/kg, and the present design would not have been economically viable.

■ Bottom line: The LCOH figures in this tool are a lower bound on actual project cost. The excluded items above — particularly STU demand charges and electricity duty — can collectively add Rs 20–40/kg or more depending on state and connection voltage.

13. Glossary

Term	Definition
AY	August Year — the Aug 22 to Aug 21 window used to define each price year.
Bid ceiling	The maximum landed power price above which the developer stops buying GDAM blocks. The number configured in the GDAM order system.
CF	Capacity Factor — actual operating hours as a fraction of 8,760 hours per year.
CRF	Capital Recovery Factor — annuity that repays a present value of 1 over N years at discount rate r: $r(1+r)^N / ((1+r)^N - 1)$.
GDAM	Green Day-Ahead Market — IEX segment for renewable-attributed power, settled at 15-minute block prices.
GH2	Green Hydrogen — hydrogen produced by electrolysis powered by non-fossil electricity.
ISTS	Inter-State Transmission System — waived for GH2 OA projects commissioned before 31 Dec 2030.
LCOH	Levelised Cost of Hydrogen — full lifecycle cost per kg of H2 produced, including capex recovery, O&M, stack replacement, power, and water.
MCP	Market Clearing Price — the exchange price at which a GDAM block clears, in Rs/MWh or Rs/kWh.
Min-run block	Minimum continuous operating duration for the electrolyser. Defaults to 2 hours (8 blocks of 15 min).
MW-invariance	Mathematical property that the LCOH-optimal ceiling and CF are independent of plant size, because all cost and output items are proportional to MW.
OA	Open Access — the right to wheel power over the grid between buyer and seller independently.
Rate base	In the repricing module, the unrecovered capital balance on which the annual return is calculated.
SEC	Specific Energy Consumption — kWh of electricity required to produce 1 kg of H2. Defaults to 52 kWh/kg for alkaline.
Sinking fund	$CRF \times NPV$ of future stack replacement expenditures, expressed as a levelised annual Rs/kg charge.
Stack replacement	Periodic replacement of the electrolyser cell stack at 60,000 operating hours (default), costing 40% of electrolyser capex.
TPA	Tonnes per annum of hydrogen produced.
Unconstrained MCP	The GDAM price that would have cleared in the absence of network congestion. Used throughout to represent the price available to a Gujarat or Rajasthan buyer on a properly evacuated connection.
Weighted average landed cost	Volume-weighted mean of the landed price across all selected blocks. The number that enters the power line of the LCOH — different from, and always below, the bid ceiling.

Table 8: Glossary of terms used in the dashboard and this document.